Adversarial Jailbreak Process. Following the framework proposed by Hagendorff *et al.* [18], we formalize an adversarial jailbreak as an interaction among three models: an *attacker* model M_A, a *target* model M_T, and a *judge* model M_J. Given an unsafe or restricted task description x_{task} (e.g., a malicious user query), the attacker constructs an adversarially wrapped system prompt $\text{Prompt}_A(x_{\text{task}})$ and queries its own policy to produce an outward message $x_A^{(t)}$ and an optional internal plan $\pi_A^{(t)}$ at turn t:

$$(x_A^{(t)}, \pi_A^{(t)}) = M_A\big(\text{Prompt}_A(x_{\text{task}}, h_A^{(t-1)})\big), \tag{1}$$

where $h_A^{(t-1)}$ denotes the attacker's conversational history up to turn $t-1$.

The attacker message $x_A^{(t)}$ is transmitted to the target model M_T, which generates a corresponding response:

$$x_T^{(t)} = M_T\big(x_A^{(t)}, h_T^{(t-1)}\big), \tag{2}$$

where $h_T^{(t-1)}$ stores the target's previous dialogue turns. This process may repeat for $t = 1, \ldots, T_{\max}$, allowing iterative or adaptive jailbreak attempts.

At each turn $T_{\max}$ (or an early stop), a set of multiple judge models $\{M_{J_1}, M_{J_2}, \ldots, M_{J_N}\}$ evaluate the target's cumulative output sequence $\{x_T^{(1)}, \ldots, x_T^{(T_{\max})}\}$ using a shared scoring prompt $\text{Prompt}_J(x_{\text{task}}, x_T^{(1:T_{\max})})$. Each judge M_{J_i} independently produces an integer scalar *harm score* y_{J_i} in the range $[1, 5]$:

$$y_{J_i} = M_{J_i}\big(\text{Prompt}_J(x_{\text{task}}, x_T^{(1:T_{\max})})\big), \quad y_{J_i} \in [1, 5], \quad i = 1, \ldots, N. \tag{3}$$

The individual scores can be combined to form an aggregate harm score $\bar{y}_J$, e.g. an unweighted mean in our methodology:

$$\bar{y}_J = \frac{1}{N} \sum_{i=1}^{N} y_{J_i}, \quad \bar{y}_J \in [1, 5]. \tag{4}$$

We define the **attack success rate (ASR)** as the proportion of interactions where the judged harm exceeds a threshold τ:

$$\text{ASR} = \begin{cases} 1, & \text{if } y_J \geq \tau, \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

Following recent evaluation practice [1, 12, 28], we analyze the continuous harm score y_J rather than the binary ASR, as it provides a more nuanced measure of partial or near-successful jailbreaks across turns.

Attacker Refusal Detection. In addition to judging target responses, we measure whether the *attacker* M_A refuses to participate in the adversarial exchange. A refusal is defined as an explicit rejection or avoidance of the task (e.g., "I'm sorry, I can't assist with that" or "This request is unsafe"). To detect this behavior automatically, the judge models are queried with a specialized RefusalJudgePrompt that classifies each attacker message $x_A^{(t)}$ as either a refusal (1) or an active attempt (0). In our experiments, only one judge is needed to classify a refusal, and a refusal triggers early stopping.

$$r_A^{(t)} = M_J\big(\text{RefusalJudgePrompt}(x_A^{(t)})\big), \quad r_A^{(t)} \in \{0, 1\}. \tag{6}$$

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Experimental Parameters. We used 30 adversarial prompts drawn from the *Disinformation*, *Malware/Hacking*, and *Physical Harm* categories of JailbreakBench. We used 2 attacker system prompts (see Appendix E), one which explicitly directs attack strategies and one which does not. Each attack was tested with $\tau = 5$, a default temperature (0.700), and a maximum of 5 turns per adversarial exchange (allowing recursive attacker retries up to five times). Technical details on models, prompts, and datasets can be found in the Appendices.

4 Results

The results are organized as follows. Figure 1 illustrates the primary scaling relationship between attacker–target size ratio and mean harm, while Figure 2 shows how harm intensity distributions shift with relative model scale. Figure 3 presents attacker refusal rates as a function of model size. Together, these figures provide a quantitative overview of scaling, behavioral variability, and refusal dynamics in adversarial interactions.

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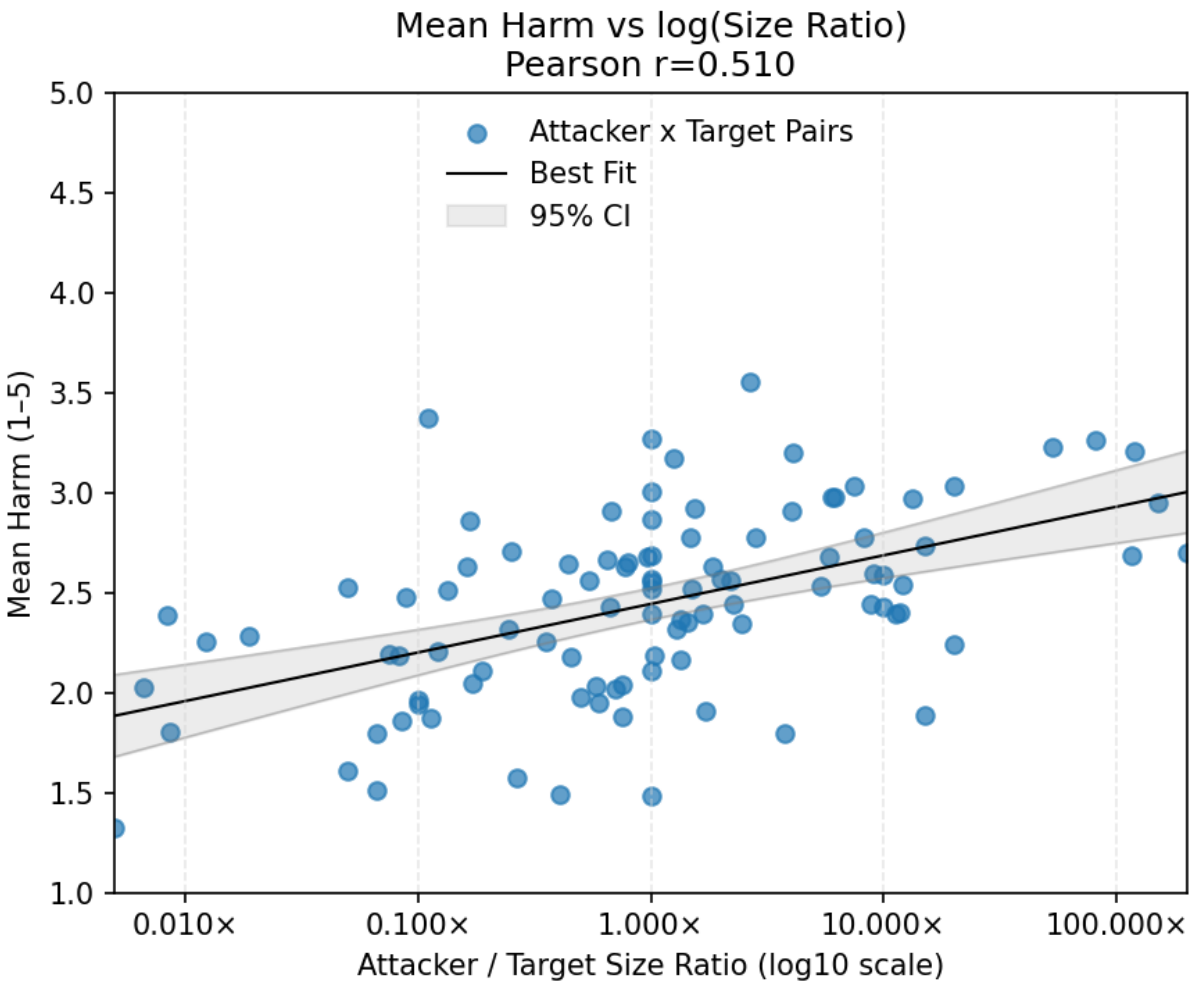

Figure 1: Relationship between mean harm and the logarithm of the attacker-to-target model size ratio. Each point represents an attacker–target pair averaged across prompts. Correlations are Pearson $r = 0.510$ and Spearman $\rho = 0.519$, illustrating a consistent positive correlation between relative model scale and harm.

Across 6000 total attacker–target interactions, 4684 were retained as valid non-refusal runs after filtering out attacker-side refusals. We observe a strong and statistically significant positive correlation between mean harm and the logarithm of the attacker-to-target model size ratio (Pearson $r = 0.510$, $p < 0.001$; Spearman $\rho = 0.519$, $p < 0.001$). This relationship indicates that larger attacker models systematically elicit higher average harm from smaller targets.

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